

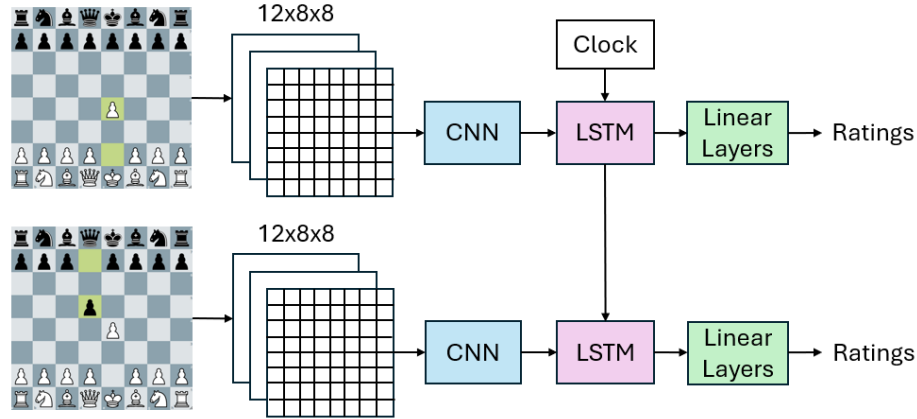


Fig. 1. The model architecture used to predict chess ratings after each move using a CNN and LSTM taking in the remaining clock time feature. This shows the first two time steps of a sample game.

Table 2. MSE of Different Methods Across Various Chess Time Controls

Time Control	RatingNet	RatingNetNoClock	Mean
Average Test Loss	56,618	91,619	183,538
Ultrabullet Loss	56,927	115,833	95,459
Bullet Loss	57,569	117,776	176,296
Blitz Loss	57,330	94,048	216,803
Rapid Loss	56,433	71,215	141,016
Classical Loss	38,491	56,032	94,304

The average MAE across the time controls is 182 for RatingNet. This is reasonable given that some chess categories, such as the United States Chess Federation (USCF), are 200 points per class [30]. The loss for longer-time controls is lower. The most likely reason for this is that the range of ratings decreases as the time control increases, partially because faster time controls are more popular. In addition, faster time controls may have a bit more randomness because players have less time to think, making it harder to estimate the skill levels.

The results may need to be taken with a grain of salt however because the percentage of times that the white player won with a higher rating and when the black player won with a higher rating was calculated only to be 53% in the training data. Such rating problems also appear in over-the-board chess games and possible factors for this center on the fact that rating systems are often overoptimistic for players with higher ratings due to the simplicity of their model [8]. Another consideration is that there is no rating label for each move, but only for the player after having played a certain number of games. The rating outputs after each move is modeling the rating estimate of the player after having seen that move and all the previous moves made by the players in that game.

We additionally report the MSE on the test set of the IEEE Big Data 2024 Cup Chess Puzzle Difficulty Competition. On the public leaderboard, the MSE was 82,049. The score is comparable to the MSE for RatingNetNoClock on estimating rating in games.

7 Clock Ablation

Because faster time controls have less time for thinking and are closer to random play than longer time controls, our hypothesis was that the correlation between move quality and chess rating may weaken, thus having additional information such as the amount of time spent on each move could be useful. To test this, an ablation experiment was performed to determine the usefulness of the clock time feature. Without the clock time feature (setting the clock input to a constant stream of 0s), the model does slightly worse; specifically, the addition of clock information reduces the MAE by 57 points, a 24% improvement. Notably, the clock feature helps significantly at faster speeds, such as bullets, with a 92-point reduction in error, or a 34% improvement. When the time control is longer, for example, in classical, the error is the lowest, with a 151 MAE for RatingNet. Overall, the network is able to utilize the time spent per move as an effective feature for skill estimation.

8 Game Analysis

A couple of sample games are investigated to determine how different types of moves affect the rating estimation of the model. We sample a game shown in Fig. 2 in which the model performs well. It correctly lowered the rating estimate of the player of the white pieces after the move Nf5, which allows the knight to be taken by the opponent’s bishop. The full game is shown in the appendix as game 1. We also analyze a game with high loss, shown in the appendix as game 2. This bullet game was played between a white rating of 2922 and a black rating of 3163. The final predicted rating for white is 1459 and for black it is 1460. Because the average rating is around 1500, we do expect higher ratings to present more difficulties because there are fewer examples of such high rated games. We expect that in some games however, RatingNet would provide more accurate rating estimations than the player’s actual rating as players do have off days or games where they play really well. With an accurate enough rating estimation system, one could consider further extending this work to potentially detect engine use or external assistance. For example, if a specific move could be marked as having a much higher rating than what the player’s true strength is, further investigation could occur.

9 Conclusion

Our proposed method for online chess rating estimation leverages game moves and clock times without relying on hand-crafted features for a variety of time con-

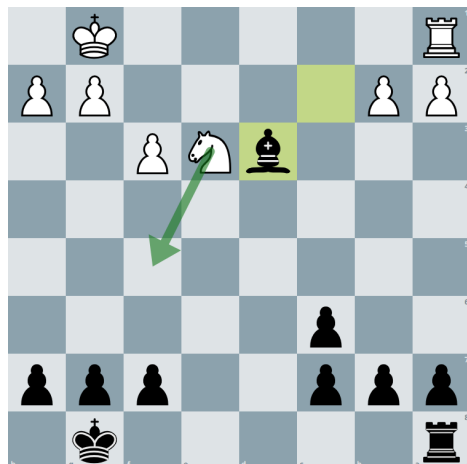


Fig. 2. White plays Nf5, a blunder because black can take it with the bishop on d3. White’s estimated rating goes from 1255 to 1241. Their actual Lichess rating in this bullet game is 1224.

trols. By integrating a CNN-LSTM architecture, we demonstrated the potential to accurately predict player rating on a move-by-move basis. The performance of the model, with an MAE of 182, shows promise in improving rating systems, providing real-time and dynamic assessments of the player’s skills. This novel approach could also serve as a foundation for applications beyond rating estimation, such as anomaly detection in games and paves the way for more granular rating systems in chess

10 Appendix

10.1 Hyperparameters and code

training batch size: 32, learning rate: 1e-4, weight decay: 1e-5, epochs: 50, learning rate patience: 10, learning rate factor: 0.5, dropout: 0.5.

The model was trained on one A40 gpu and took 12 hours to finish training. The code: <https://github.com/AstroBoy1/RatingNet>

10.2 Game 1

1. e4 e5 2. Nf3 Nc6 3. d4 exd4 4. Nxd4 Nf6 5. Nxc6 dxc6 6. Nc3 Qxd1 7. Nxd1 Bc5 8. Bd3 O-O 9. O-O Re8 10. Be3 Bxe3 11. Nxe3 Nxe4 12. Bxe4 Rxe4 13. Nd1 Bf5 14. f3 Re2 15. Rf2 Rxc2 16. Rxc2 Bxc2 17. Ne3 Bd3 18. Nf5 Bxf5 19. g4 Bg6 20. Kf2 Rd8 21. Kg3 Rd2 22. Rb1 Bxb1 23. h4 Rxb2 24. h5 Rxa2 25. g5 Bf5 26. f4 h6 27. Kh4 hxg5 28. Kxg5 Bd3 29. h6 gxh6 30. Kxh6 Rg2 31. f5 f6 32. Kh5 Be2 33. Kh6 Rg5 1/2-1/2